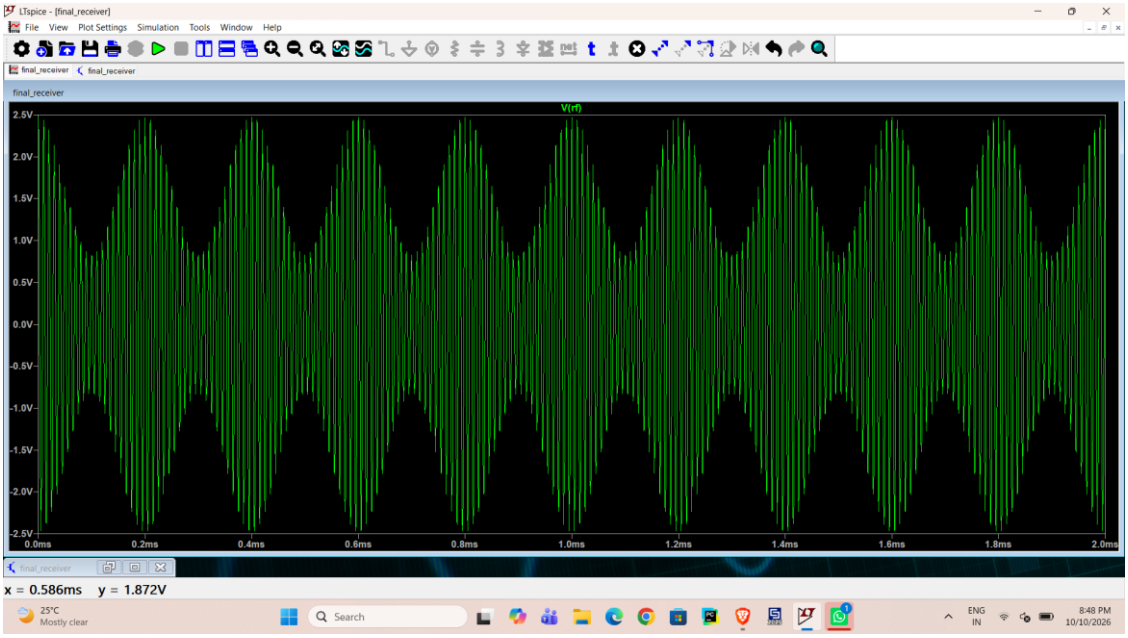
Results of Each Stage

Stage 1 (the two-stage op-amp) shows Cadence Virtuoso / Spectre results; its sizing and schematic are in 03_circuits_and_design.docx. All other plots are LTspice results from the final receiver design. Numbers marked *ngspice* were checked with an independent ngspice run of the same circuit. Numbers marked *plot* are read from the screenshot and are approximate.

Stage 0: RF input signal (AM)



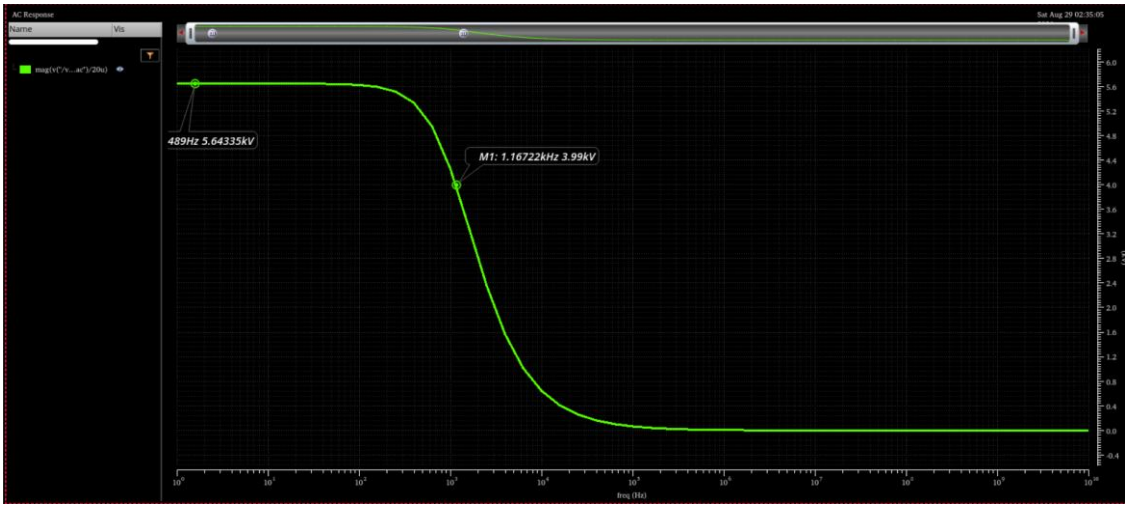
RF AM signal

What it shows

- The 100 kHz carrier is amplitude-modulated by the 5 kHz message: the envelope repeats every 0.2 ms.
- The envelope peaks at about 2.475 V and its valleys are at about 0.825 V (*plot*), which gives a modulation index $\mu = (2.475 - 0.825)/(2.475 + 0.825) = 0.5$.
- The signal is centered on 0 V, as an antenna signal would be. The mixers add the DC offsets later.
- This is the input of the receiver. The goal is to recover the 5 kHz envelope information from it.

Stage 1: Two-stage CMOS op-amp (Cadence Spectre)

1a. Open-loop AC gain and bandwidth

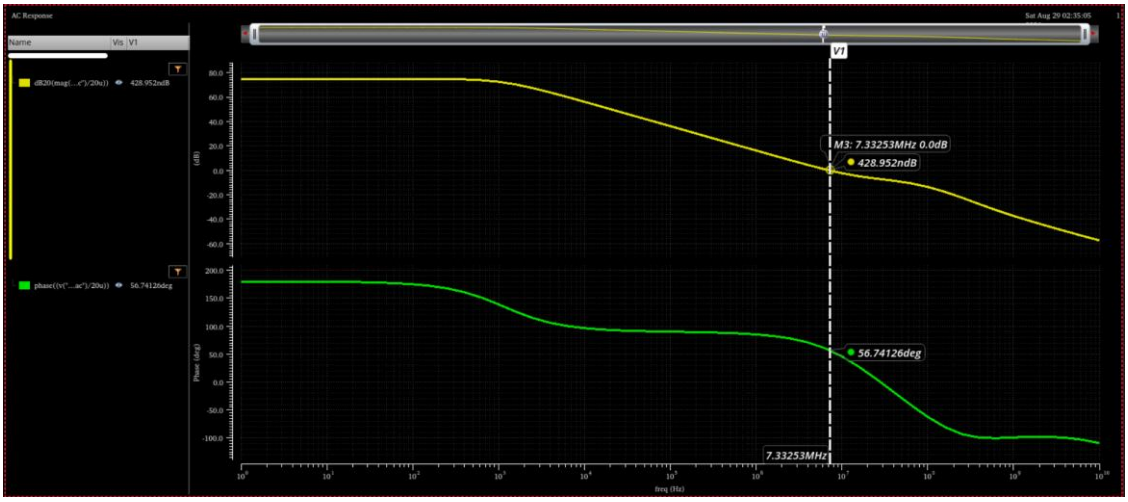


Open-loop differential AC response, mag(vout)/vin. Low-frequency plateau = 5.64335 kV/V; −3 dB point (3.99 kV/V) at f = 1.16722 kHz.

Parameter	Value
DC open-loop gain, A_v,dc	5643.35 V/V (75.03 dB)
−3 dB bandwidth, f−3dB	1.16722 kHz
Gain-bandwidth product, A_v,dc × f−3dB	≈ 6.587 MHz
Hand-calculated gain (gm/Id)	≈ 6123 V/V (75.74 dB), 8.5 % above simulation

The gain is flat at about 75 dB at low frequency and then rolls off at 20 dB per decade after the dominant pole set by the Miller capacitor. The design target of 5000–6000 V/V is met. Check at the −3 dB point: 5643.35/√2 = 3990.5 V/V.

1b. Loop gain and phase margin

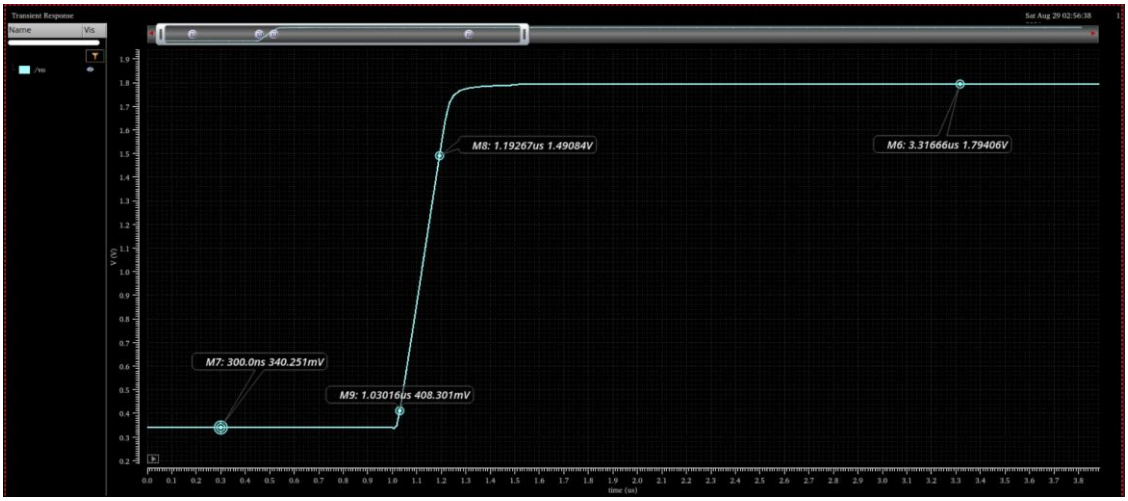


Loop-gain AC response. Top: magnitude, dB20(mag(vloop)). Bottom: phase. Marker at the 0 dB crossover.

Parameter	Value
Unity loop-gain crossover, f0dB	7.33253 MHz
Phase at f0dB	56.74126°
Phase margin, PM	≈ 56.74°

A phase margin of about 56.7° lies inside the usual 45°–60° guideline, so a step input should show one lightly damped overshoot and no sustained ringing, consistent with the transient plots below.

1c. Large-signal transient and slew rate

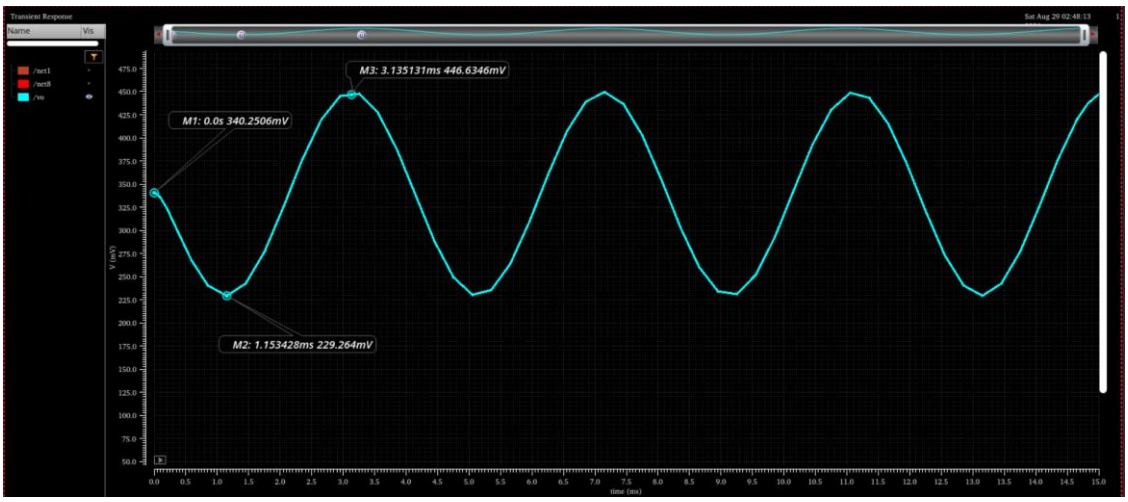


Large-signal transient output /vo for a full-scale input step. Markers on the rising edge give the slew-rate window.

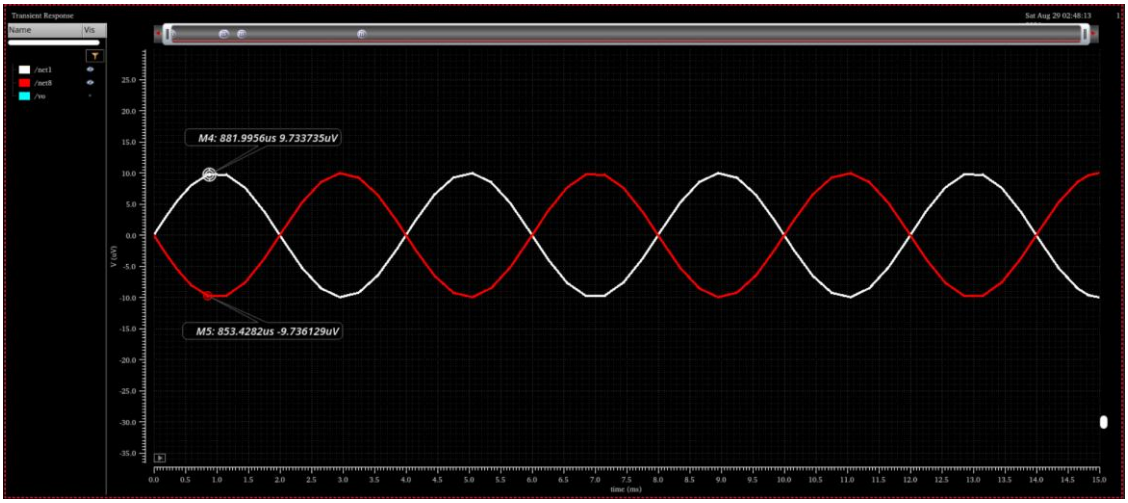
Parameter	Value
Pre-step baseline	340.251 mV at t = 300 ns
Rising-edge marker 1	408.301 mV at t = 1.03016 µs
Rising-edge marker 2	1.49084 V at t = 1.19267 µs
Final settled output	1.79406 V at t = 3.31666 µs (99.67 % of VDD)
Slew rate, SR = ΔV/Δt	(1.49084 − 0.408301) V / (1.19267 − 1.03016) µs ≈ 6.661 V/µs
Large-signal settling time	≈ 2.317 µs (from the ≈ 1 µs edge to 99.67 % of VDD)

1d. Small-signal differential gain, time-domain check

A low-frequency differential sine was applied at the Stage-1 inputs (net1, net8) and the output /vo captured over several cycles, as a time-domain cross-check of the AC gain.



Output waveform /vo. Peak = 446.6346 mV, trough = 229.264 mV.



Differential input stimulus. net1 peak = 9.733735 µV; net8 trough = −9.736129 µV (equal magnitude, opposite phase).

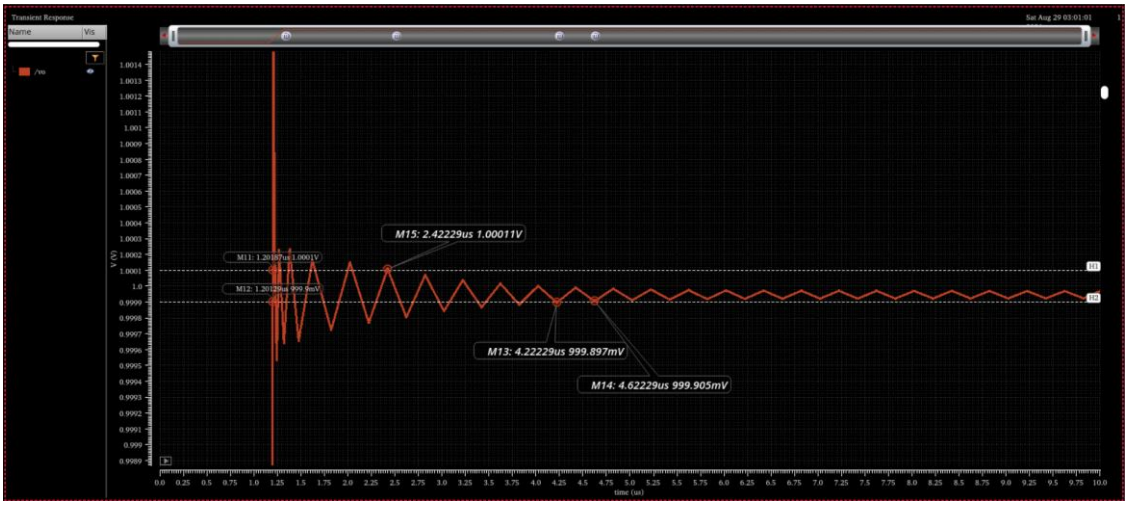
Parameter	Value
Vout peak / trough	446.6346 mV / 229.264 mV
Vout peak-to-peak	217.3706 mV
Vin,diff amplitude (|v_net1| + |v_net8|)	19.4699 µV
Vin,diff peak-to-peak	38.9397 µV
Measured gain	217.3706 mV / 38.9397 µV ≈ 5582.2 V/V
Comparison with AC open-loop gain	5643.35 V/V; deviation ≈ 1.08 %

1e. Fine-resolution step response

A 0 → 1 V step was captured at fine time resolution to look at settling around the final value, with a ±0.01 % band (1.0001 V and 0.9999 V) overlaid.



Full 0 → 1 V step response /vo with the ±0.01 % band. Rising edge and early settling markers M10–M14.



Zoomed view of the previous plot, showing the settling markers M10–M15.

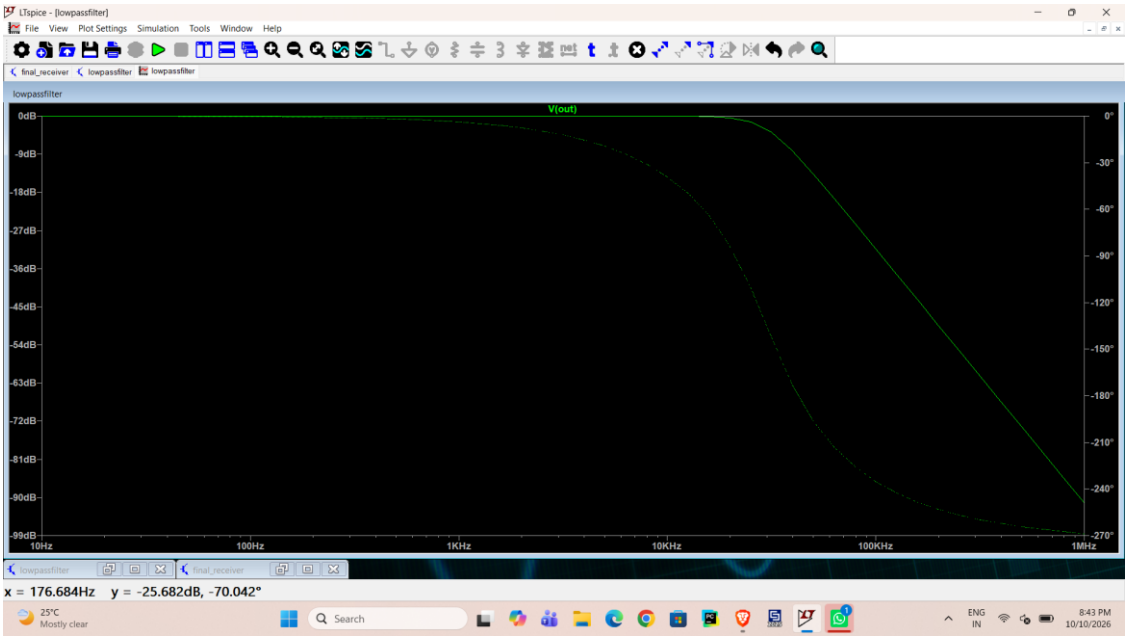
Marker	Value
M10 (rising edge)	1.00185 V at t = 1.20694 µs, within 0.19 % of the final value
M15	1.00011 V at t = 2.42229 µs
M13	999.897 mV at t = 4.22229 µs
M14	999.905 mV at t = 4.62229 µs

The output reaches within 0.2 % of the final value about 1.2 µs after the step starts. The samples from 2.42 µs onward lie within ±0.011 % of 1 V, so the response is settled to roughly ±0.01 % by about 2.4 µs.

Summary of Stage 1. The op-amp is a well-damped, single-dominant-pole, Miller-compensated amplifier: 75.03 dB DC gain, 6.59 MHz gain-bandwidth, 56.7° phase margin and 6.66 V/µs slew rate, at about 96 µA from 1.8 V.

> The filter and receiver stages below (Stages 2 to 4) are the LTspice results from receiver simulation in LTspice , run with the op-amp model on a 3.3 V supply. They have been simulated with the Cadence op-amp design.

Stage 2: 3rd-order Butterworth low-pass filter (AC response)

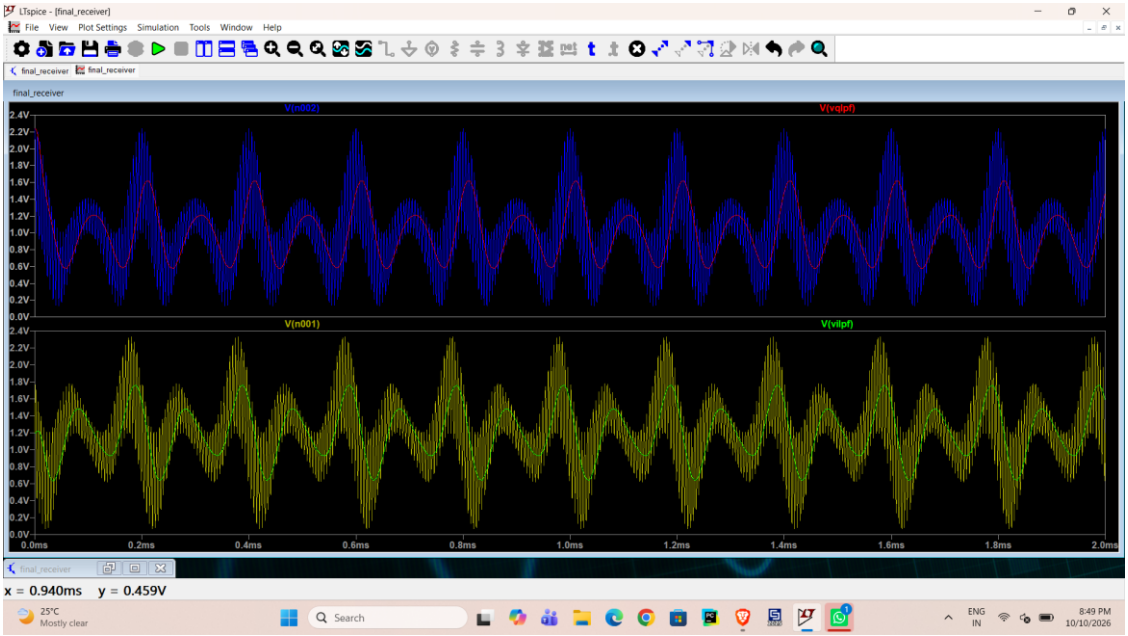


Low-pass filter AC response

What it shows

- The gain is flat at 0 dB in the passband and the **–3 dB frequency is 30 kHz** (29.9 kHz).
- Gain at the frequencies of interest :10 kHz = 0.995, 15 kHz = 0.988 (message sidebands pass with about 0.1 dB loss), 100 kHz = 0.027 (−31 dB), 190 kHz = 0.004 (−48 dB).
- The roll-off is 60 dB per decade, reaching about −91 dB at 1 MHz (*plot*), which is the 3rd-order slope. The phase goes towards −270 degrees (three poles), with about −39 degrees at the 10 kHz IF.
- The 190 kHz mixer sum product is removed, while the 10 kHz IF and its 5 kHz sidebands pass.

Stage 3: Mixer outputs and filtered I/Q signals

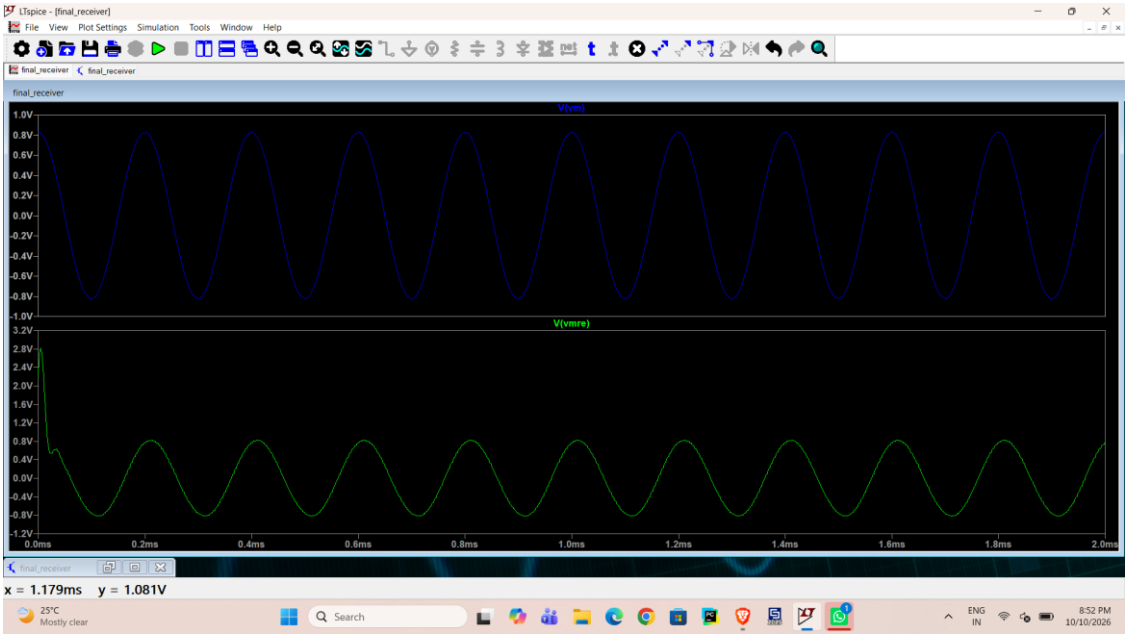


Mixer outputs and filtered I/Q

What it shows

- Top panel is the Q path and bottom panel is the I path. The dense traces (`V(n002)`, `V(n001)`) are the mixer outputs before filtering: they swing about 0.1 V to 2.35 V and carry fast 190 kHz ripple on top of the wanted signal.
- The smooth traces (`V(vqlpf)`, `V(vilpf)`) are the low-pass outputs: the ripple is gone and only the 10 kHz IF remains, with the 5 kHz envelope (two IF cycles per envelope cycle).
- DC levels are as designed: I is centered at 1.2 V (range about 0.6 to 1.75 V) and Q around 1.0 V (range about 0.6 to 1.6 V) (*plot*), so all signals stay inside the 0 to 3.3 V supply range.
- I and Q have different shapes because they are the sine and cosine components of the same IF, which is what the second mixer needs.

Stage 4: Recovered message compared with the original



Message vs recovered

What it shows

- Top: the original message `V(vm)`, a 5 kHz sine of about ±0.82 V. Bottom: the recovered signal `V(vmre)`, also a clean 5 kHz sine of about ±0.83 V centered on 0 V (*plot*).
- Amplitude and frequency match, so the carrier, the IF and the DC offsets are all removed correctly.
- The recovered wave lags the original by about 10 µs, which matches the filter group delay (≈ 2/ω_c = 10.6 µs).
- The spike near t = 0 (up to about 2.8 V) is the start-up transient of the filters and settles within about 0.1 ms.

Summary

Stage	Key result
RF input	AM, μ = 0.5, envelope 0.825 V to 2.475 V
Op-amp	DC gain 75.03 dB (5643 V/V), GBW ≈ 6.59 MHz, PM ≈ 56.7°, SR ≈ 6.66 V/μs, ≈ 96 μA at 1.8 V
Low-pass filter	−3 dB at 29.9 kHz, −48 dB at 190 kHz, 60 dB/decade
I/Q after filter	10 kHz IF with 5 kHz envelope, 190 kHz ripple removed
Recovered message	about ±0.83 V, 5 kHz, delay about 10 μs